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Effects of Coconut Shell Ash and Coir Fiber on the Mechanical Properties and Microstructure of Concrete

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Abstract

The utilization of cement is one of the primary sources of carbon emissions in concrete, driving the search for sustainable alternative materials. Although extensive research has been conducted on the use of agricultural waste as supplementary cementitious materials (SCMs), the effects of coconut shell ash (CSA) and coir fiber (CF) on concrete properties have not been extensively investigated. This study systematically investigates the influence of CSA as a SCM (0–20%) and CF as a reinforcement material (0–0.32%) on the workability, density, compressive strength, flexural strength, splitting tensile strength, and failure modes of concrete, complemented by microstructural mechanism analysis. The cement and CSA were characterized using XRF, XRD, and SEM. The results indicate that the incorporation of both CSA and CF reduces the workability and density of concrete. For concrete with CSA only, the compressive strength decreases by up to 24.7% when the replacement level reaches 20%. However, concrete with 10% CSA still maintains 87.2% of the strength of ordinary concrete, which satisfies the C40 requirement. In contrast, CF incorporation alone improves the mechanical properties, with compressive strength, flexural strength, and splitting tensile strength reaching peak increases of 6.4%, 13.9%, and 7.5%, respectively, when the CF content is 0.24%. Incorporating 0.16% CF into 10% CSA concrete mitigates the strength reduction caused by CSA, achieving compressive, flexural, and splitting tensile strengths of 47.99 MPa, 5.63 MPa, and 3.99 MPa, respectively (95.7%, 98.3%, and 96.4% of the strengths of ordinary concrete). Microstructural analysis reveals that CSA deteriorates the interfacial transition zone (ITZ), while CF compensates for partial strength loss through the bridging effect, although its reinforcement efficiency is influenced by fiber dispersion and ITZ quality. This study provides a theoretical foundation and technical reference for the utilization of coconut shell waste in sustainable concrete.

Keywords: coconut shell ash; coir fiber; sustainable concrete; interfacial transition zone (ITZ); fiber bridging effect

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1. Introduction

Concrete, as one of the most widely used foundational materials in modern construction engineering, occupies a central position in infrastructure development, including buildings, bridges, and roads, owing to its excellent mechanical properties and durability. However, the production of traditional concrete is highly dependent on non-renewable resources such as cement and natural aggregates. Notably, cement manufacturing generates substantial carbon dioxide emissions due to high-temperature calcination, accounting for approximately 7% of global anthropogenic CO_2 emissions [1,2], posing significant environmental pressure. To address this challenge and promote the low-carbon transition of construction materials, many researchers are actively exploring eco-friendly and sustainable alternative materials to mitigate the environmental impact of the concrete industry [3–5].

In recent years, industrial and agricultural wastes possessing pozzolanic activity or physical reinforcement characteristics have been extensively investigated as supplementary cementitious materials (SCMs) [6–8]. Among these, industrial by-products such as fly ash [9], silica fume [10], and slag [11] have achieved large-scale application. Agricultural wastes, including rice husk ash [12,13], walnut shell ash [14], cotton stalk ash [15], and coconut shell ash (CSA) [16] also demonstrated potential as SCMs. CSA, produced by calcining and grinding coconut shells, contains SiO_2 and Al_2O_3 and exhibits pozzolanic activity [17]. Joshua et al. [18] found that the strength activity index of the cement mortar with 20% CSA replacing cement was 92% (greater than 75%), leading to the inference that CSA possesses pozzolanic activity. Saraswat et al. [19] found that when CSA replaced 0–25% of cement, the compressive and splitting tensile strengths of concrete initially increased and then decreased with the replacement ratio, peaking at 10%. P Vasanthi et al. [20] found that replacing 12% of the cement with CSA improved the flexural strength of the concrete. Priya et al. [21] reported that concrete with 12.5% cement replacement by CSA exhibited a 43.3% higher 28-day compressive strength than plain concrete, with the enhancement increasing with longer curing ages. Ranatunga et al. [22] observed that replacing 15% of cement with CSA enhanced concrete strength, but further increases led to a gradual decline, and the incorporation of CSA significantly reduced slump. Bheel et al. [23] reported that a 10% CSA replacement not only enhanced the mechanical properties but also reduced the embodied carbon of the concrete. Conversely, other studies indicate that incorporating CSA reduces concrete strength [24,25]. Rilwan et al. [26] reported that both compressive and flexural strengths decreased progressively as the CSA replacement ratio increased from 0% to 15%. Lawrence et al. [27] found that increasing the CSA content not only reduced compressive strength but also decreased the bulk density. Adajar et al. [28] discovered that concrete with 10% CSA achieved 92.10% of the compressive strength of ordinary concrete, with further reductions at 20–40% replacement, accompanied by significant changes in failure modes. Although existing studies present discrepancies regarding the influence of CSA on concrete strength, the general consensus is that the replacement ratio should not exceed 10–15% to avoid adverse effects on mechanical performance. In addition, other researchers have found that the incorporation of CSA can enhance the durability properties of concrete [29]. Kumar et al. [30] reported that adding 10% CSA reduced the water absorption of concrete, thereby improving its durability. Hanumanthappa [31] et al. found that concrete containing 20% CSA exhibited a 35.3% reduction in total charge passed during the rapid chloride permeability test, indicating enhanced resistance to chloride ion penetration.

Plant fibers are widely utilized to improve concrete properties due to their renewability and mechanical enhancement effects [32–34]. Coir fiber (CF), another significant component of coconut waste, is often employed as a reinforcing material to enhance concrete performance owing to its high tensile strength and good toughness [35,36]. Ali et al. [37] found that CF with a length of 5 cm and a dosage of 5% yielded the best improvement in the mechanical properties of concrete. Shcherban et al. [38] found that the optimal dosage of CF was 1.75%, at which point the mechanical properties of normal-weight concrete, such as strength and ultimate strain, were significantly enhanced. Research indicate that incorporating an appropriate amount of CF can optimize mechanical properties. Katman et al. [39] found that incorporating 0%, 1%, 2%, 3%, and 4% CF (by concrete volume) resulted in strengths that initially increased and then decreased with fiber content. Ahmad et al. [40] reported that incorporating 0.5%, 1%, 1.5%, and 2% CF (by cement mass) led to an initial increase followed by a decrease in concrete strength, with peak strength at 1.5% CF; further increases caused fiber agglomeration and strength reduction. Furthermore, studies suggest

CF can provide a "performance compensation" effect in concrete containing SCMs or waste materials. Alomayri et al. [41] reported that replacing 25% of cement with ground steel slag reduced compressive strength by 7%, but incorporating 0.25% CF (by concrete volume) subsequently increased strength by 8% compared to plain concrete. Silva et al. [42] found that incorporating 0.5% CF mitigated the reduction in compressive and flexural strengths caused by replacing 10% of cement with rice husk ash. Babar Ali et al. [43] reported that the use of 2% CF and a high-range water reducer helped mitigate the negative impact of recycled aggregates on compressive strength, resulting in improvements of 9% and 12% for concrete with 50% and 100% recycled aggregate replacement ratios, respectively. Our previous study [44] also indicated that in concrete with 10% coconut shell replacing natural aggregate, incorporating 0.24% CF (by concrete mass) restored strength to the level of plain concrete. The influence of CF on the failure mode of concrete is also significant. Hwang et al. [45] discovered that incorporating CF had positive effects on both the plastic shrinkage cracking and impact resistance of cement-based composites. Das et al. [46] found that the incorporation of CF changed the failure crack pattern of cylindrical concrete specimens from vertical cracks to wedge-shaped cracks; furthermore, the crack length increased with higher fiber content.

In summary, partially replacing cement with CSA reduces resource consumption and carbon emissions, but high replacement ratios easily lead to strength degradation. CF not only enhances the mechanical properties of ordinary concrete but also effectively compensates for the strength reduction caused by waste incorporation. This study aims to develop eco-friendly concrete by partially replacing cement with CSA and incorporating CF to offset potential mechanical losses, achieving performance comparable to conventional concrete. The research first characterized the physical, chemical, and microstructural properties of cement and CSA. Subsequently, concrete mixtures with varying CSA replacement ratios and CF contents were designed and prepared. By testing the slump, density, compressive strength, flexural strength, and splitting tensile strength, combined with microstructural observations, the mechanisms by which CSA and CF influence concrete performance were systematically analyzed.

2. Materials and Methods

This study was conducted in three main stages: (1) the processing and preparation of raw materials; (2) the preparation and curing of concrete; and (3) the testing of the concrete's properties. The research framework is illustrated in Figure 1.

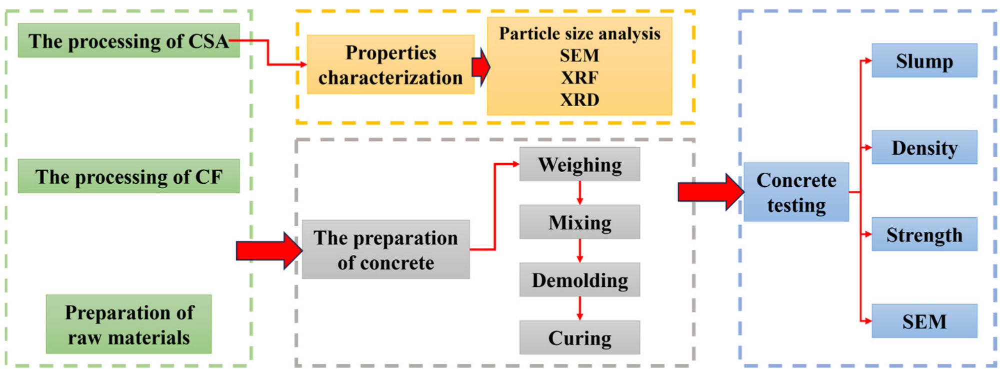

Figure 1. The research framework.

2.1. Raw Materials

The materials employed in this study for concrete fabrication encompass cement, coconut shell ash (CSA), coir fiber (CF), granite gravel, river sand, water, and admixtures.

The cement utilized is ordinary Portland cement with a strength class of 42.5. The waste CSA was collected from a coconut shell activated carbon manufacturing plant in Hainan. Its processing procedure is as follows: initially, the raw CSA was dried in an oven at 105 °C to constant weight; subsequently, it was ground using a grinder; finally, coarse particles were sieved out using a 75 μm sieve. The CSA before and after treatment is depicted in Figures 2a and 2b, respectively. The CF was sourced from a coconut food processing factory in Hainan. The treatment procedure is as follows: firstly, the fiber bundles were cut into short fibers of 20 mm in length; then, the CF was immersed in a 5% NaOH solution for 2 h to remove surface impurities and enhance its interfacial bonding with the cement matrix [47–49]; following this, residual alkali was neutralized with acetic acid and the fibers were rinsed repeatedly with clean water until neutral; finally, the CF was dried in an oven at 60 °C for 24 h. The processed CF is illustrated in Figure 2c.

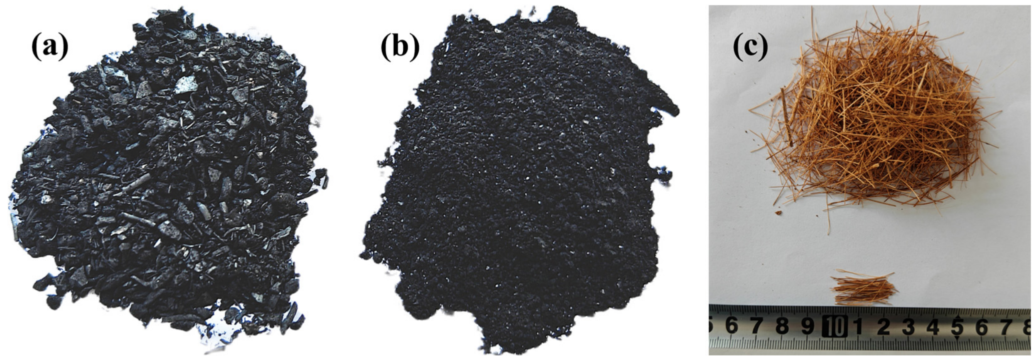

Figure 2. Raw materials: (**a**) the original CSA, (**b**) processed CSA, and (**c**) CF.

The coarse aggregate used in the experiment is granite crushed stone, with a crushing value of 8.83%, satisfying the requirements for Class I crushed stone. The fine aggregate is river sand sourced locally in Hainan. The bulk density, water absorption, and particle grading of the aggregates were determined according to GB/T 14684-2022 [50] and GB/T 14685-2022 [51], respectively. The particle grading curves for the granite gravel and river sand are presented in Figure 3, and their physical properties are summarized in Table 1. Tap water was used in the experiments. The chemical admixture employed is polycarboxylate high-performance water reducer.

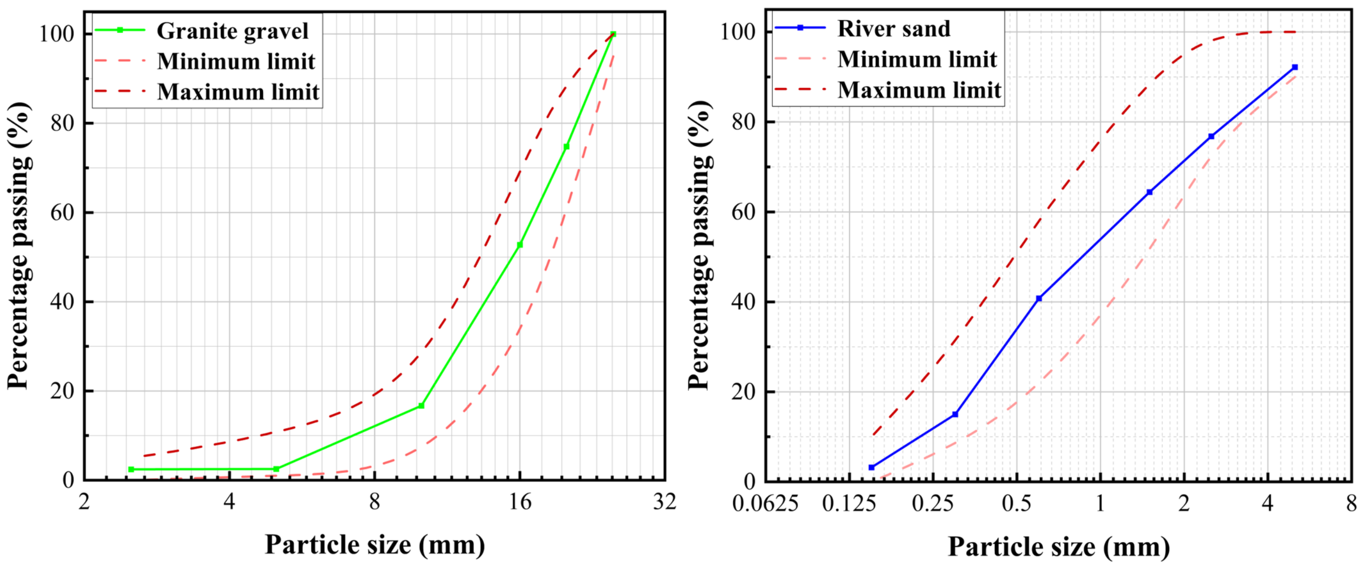

Figure 3. The particle gradation of granite gravel and river sand.

Table 1. Material properties of crushed stones and river sand.

Material Property	Granite Gravel	River Sand
Bulk density (kg/m^3)	1557	1619
Absorption (%)	3.38	2.36
Fineness Modulus	—	2.75
Crushing Index (%)	8.83	—

2.2. Preparation of Concrete

Based on C40 concrete, thirteen mix proportions were designed in this study, with detailed data presented in Table 2. In these mixes, CSA was used to replace cement at 0, 5, 10, 15, and 20% by mass, and CF was added at 0, 0.08, 0.16, 0.24, and 0.32% by mass of concrete (corresponding to 0%, 0.33%, 0.67%, 1%, and 1.33% by volume, respectively). Each mix design included 9 cubic specimens (100 mm sides) and 3 prismatic specimens (100 mm × 100 mm × 400 mm) for performance testing, totaling 156 specimens.

Table 2. Mix proportions of concrete (kg/m^3).

Mix ID	Cement	CSA	Granite Gravel	River Sand	CF	Water	Admixture
CSA0CF0 *	400	0	940	960	0	150	5
CSA5CF0	380	20	940	960	0	150	5
CSA10CF0	360	40	940	960	0	150	5
CSA15CF0	340	60	940	960	0	150	5
CSA20CF0	320	80	940	960	0	150	5
CSA0CF8	400	0	940	960	1.96	150	5
CSA0CF16	400	0	940	960	3.93	150	5
CSA0CF24	400	0	940	960	5.89	150	5
CSA0CF32	400	0	940	960	7.86	150	5
CSA10CF8	360	40	940	960	1.96	150	5
CSA10CF16	360	40	940	960	3.93	150	5
CSA10CF24	360	40	940	960	5.89	150	5
CSA10CF32	360	40	940	960	7.86	150	5

Note. * CSA0CF0 denotes plain concrete without CSA and CF. CSA10CF32 indicates that the binder consists of 90 wt% cement and 10 wt% CSA, with 0.32 wt% (by mass of concrete) CF incorporated.

During preparation, the raw materials were weighed and introduced into a mixer for uniform blending. It should be noted that the CF was added in small quantities and multiple batches to prevent fiber entanglement. After mixing, the mixture was cast into corresponding molds. Following 24 h of casting, the specimens were cured for 28 d in a standard curing room at a temperature of 20 °C and a relative humidity of over 95% after demolding.

2.3. Testing Methods

The slump of each concrete mixture was measured according to GB/T 50080-2016 [52]. The density was determined via the drying method in accordance with GB/T 50081-2019 [53]. All mechanical property tests were conducted in compliance with GB/T 50081-2019. Compressive strength tests were performed on 100 mm × 100 mm × 100 mm cubic specimens using a WAW-1000 universal testing machine (Shanghai Longhua Testing Instrument Co., Ltd., Shanghai, China) at a loading rate of 0.6 MPa/s. Flexural strength and splitting tensile strength tests were conducted using a WAW-100 universal testing machine. For the flexural test, a three-point bending method was employed with specimens of size 100 mm × 100 mm × 400 mm, a support span of 300 mm, and a loading rate of 0.06 MPa/s. For the splitting tensile strength test, 100 mm × 100 mm × 100 mm cubic specimens were placed in a positioning fixture at a loading rate of 0.06 MPa/s. The configurations for the mechanical tests are illustrated in Figure 4.

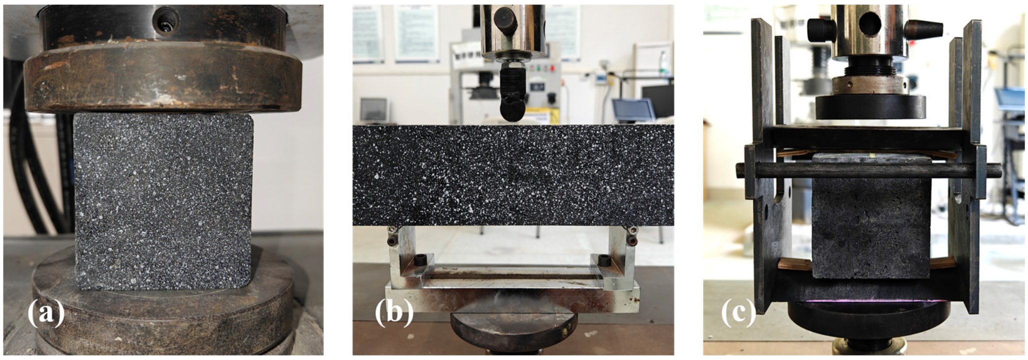

Figure 4. (**a**) Compressive strength test, (**b**) flexural strength test, and (**c**) splitting tensile strength test.

To characterize the fundamental properties of the cement and CSA, their chemical compositions were determined using a Rigaku ZSX Primus III+ X-ray fluorescence (XRF) spectrometer (Rigaku Holdings Corporation, Tokyo, Japan). Phase compositions were analyzed using a Rigaku Ultima IV X-ray diffractometer (XRD; Rigaku Holdings Corporation, Tokyo, Japan). Particle size distributions were measured using a BT-9300H laser particle size analyzer (Dandong Bettersize Instruments Ltd., Dandong, Liaoning Province, China). Particle morphologies were observed using a ZEISS Sigma 360 scanning electron microscope (SEM; Carl Zeiss AG, Auberkhain, Germany). Furthermore, to investigate the influence of CSA and CF on the microstructure of concrete, SEM observations were also conducted on the concrete matrix.

3. Results and Discussion

3.1. Properties of Cement and CSA

The particle size distribution curves of cement and CSA are presented in Figure 5, and their physical properties are listed in Table 3. The results indicate that the particle size of CSA is generally smaller than that of cement. The median particle diameter (D50) of CSA is 9.57 μm, whereas that of cement is 19.12 μm. Furthermore, particles smaller than 45 μm account for 99.97% of CSA and 85.67% of cement. The finer particles of CSA contribute to the effective filling of internal voids in concrete, thereby enhancing its filler effect. The micro-morphologies of cement and CSA observed via SEM are shown in Figures 6 and 7. It is evident that cement particles are relatively uniform in size with smoother surfaces, whereas CSA particles are irregular and rough with abundant micropores. These surface characteristics result in a larger specific surface area and stronger water absorption capacity for CSA compared to cement.

Table 3. The physical properties of cement and CSA.

Property	Cement	CSA
Diameter of the particle corresponding to 90% finer (μm)	52.06	22.92
Diameter of the particle corresponding to 50% finer (μm)	19.12	9.57
Diameter of the particle corresponding to 10% finer (μm)	4.17	2.75
Portion of the particles smaller than 45.5 μm (%)	85.67	99.97
Specific surface area (m^2/kg)	326.2	396.9
Bulk density (kg/m^3)	1234.75	639.72

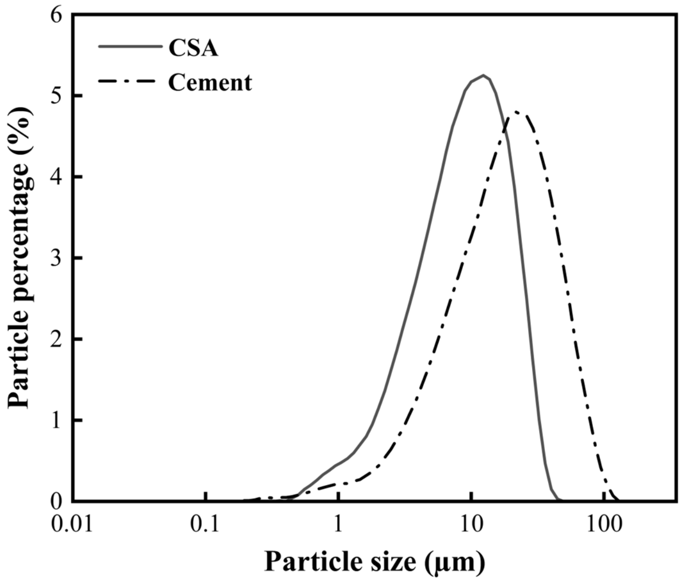

Figure 5. The particle size distribution of cement and CSA.

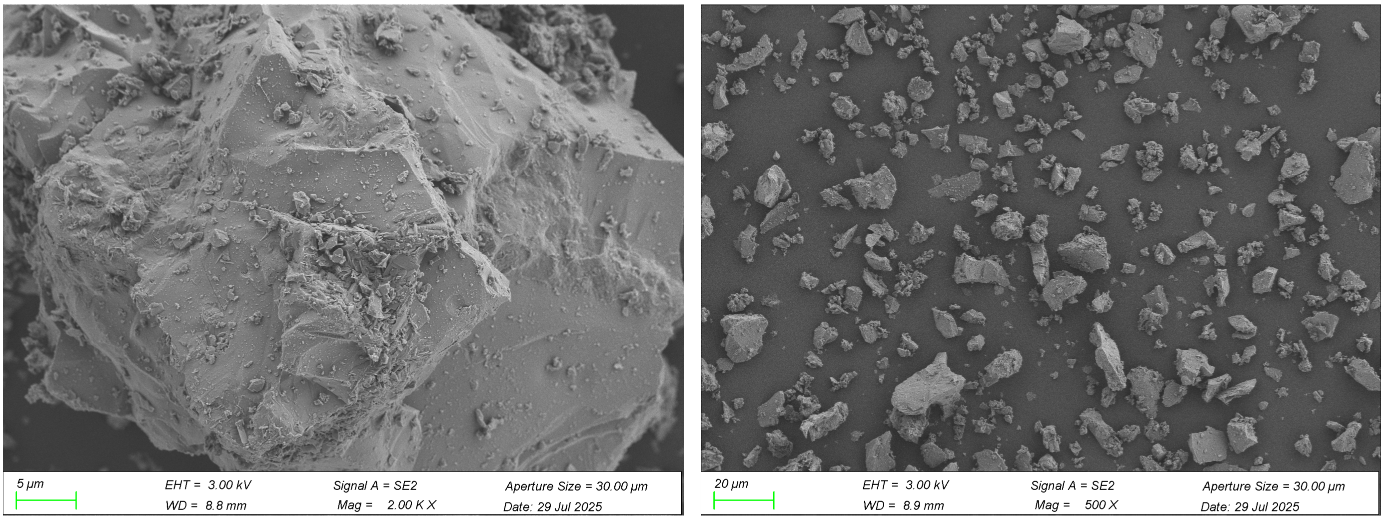

Figure 6. SEM images of cement.

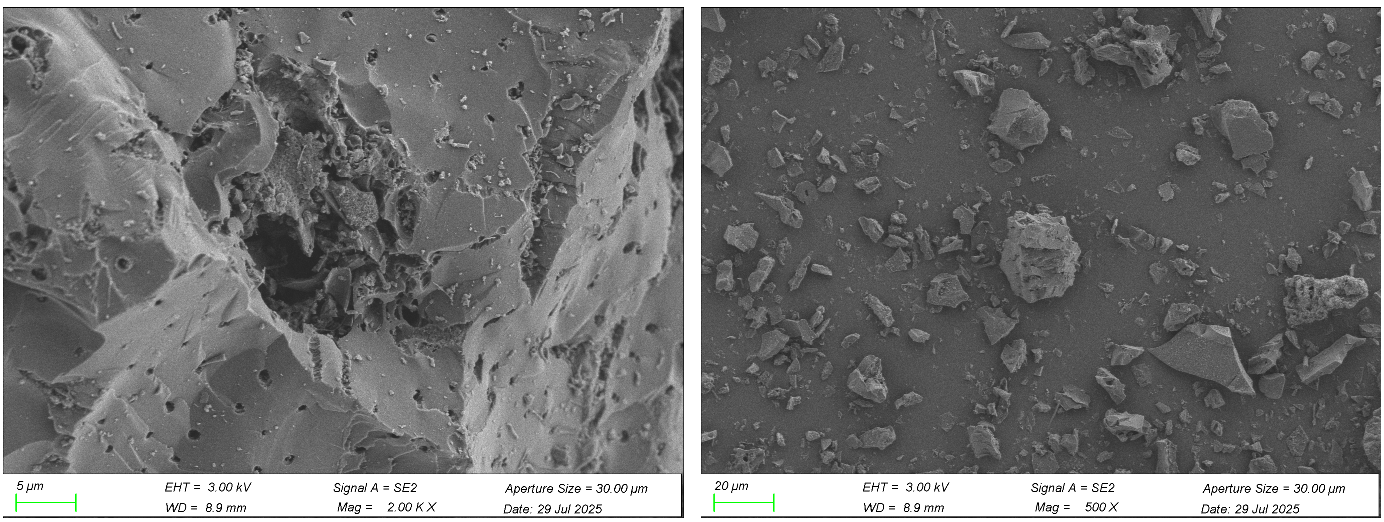

Figure 7. SEM images of CSA.

The chemical compositions and XRD diffractograms of cement and CSA are detailed in Table 4 and Figure 8, respectively. The XRF results reveal that the primary chemical components of cement are CaO, SiO_2, Al_2O_3, and Fe_2O_3. During cement hydration, CaO and SiO_2 participate in the formation of C-S-H gel and $Ca(OH)_2$, where the C-S-H gel serves as the primary source of concrete strength [54,55]. The main chemical components of CSA are K_2O, CaO, SiO_2, and Fe_2O_3. The pozzolanic reaction between CSA and $Ca(OH)_2$ generated from cement hydration produces C-S-H gels. Distinct characteristic diffraction peaks of tricalcium silicate ($3CaO{\cdot}SiO_2$), dicalcium silicate ($2CaO{\cdot}SiO_2$), tricalcium aluminate ($3CaO{\cdot}Al_2O_3$), and tetracalcium aluminoferrite ($4CaO{\cdot}Al_2O_3{\cdot}Fe_2O_3$) are observed in the XRD diffractogram of cement, indicating good crystallinity. In the XRD diffractogram of CSA, multiple characteristic diffraction peaks of SiO_2 are detected, suggesting the presence of crystalline SiO_2. The existence of two broad diffraction peaks at $2\theta \approx 20$–$26°$ and $\theta \approx 40$–$45°$ suggests the CSA contains amorphous carbon. The appearance of KCl diffraction peaks is primarily attributed to the high potassium content inherent in coconut shells, which combines with chlorine during combustion to form residual KCl crystals in the ash. The presence of KCl may pose long-term durability risks. Additionally, characteristic calcite ($CaCO_3$) peaks observed at $2\theta = 29°$ and $39°$ are likely secondary minerals formed by the absorption of CO_2 from the air by the CSA.

Table 4. Chemical composition of cement and CSA.

Compound	Cement (%)	CSA (%)
CaO	58.31	23.26
SiO_2	21.72	20.62
Al_2O_3	7.71	1.41
Fe_2O_3	4.39	13.48
SO_3	2.81	2.14
MgO	2.67	1.28
K_2O	0.82	30.61
Cl	0.04	2.14
P_2O_5	0.09	2.09
LOI	—	2.87

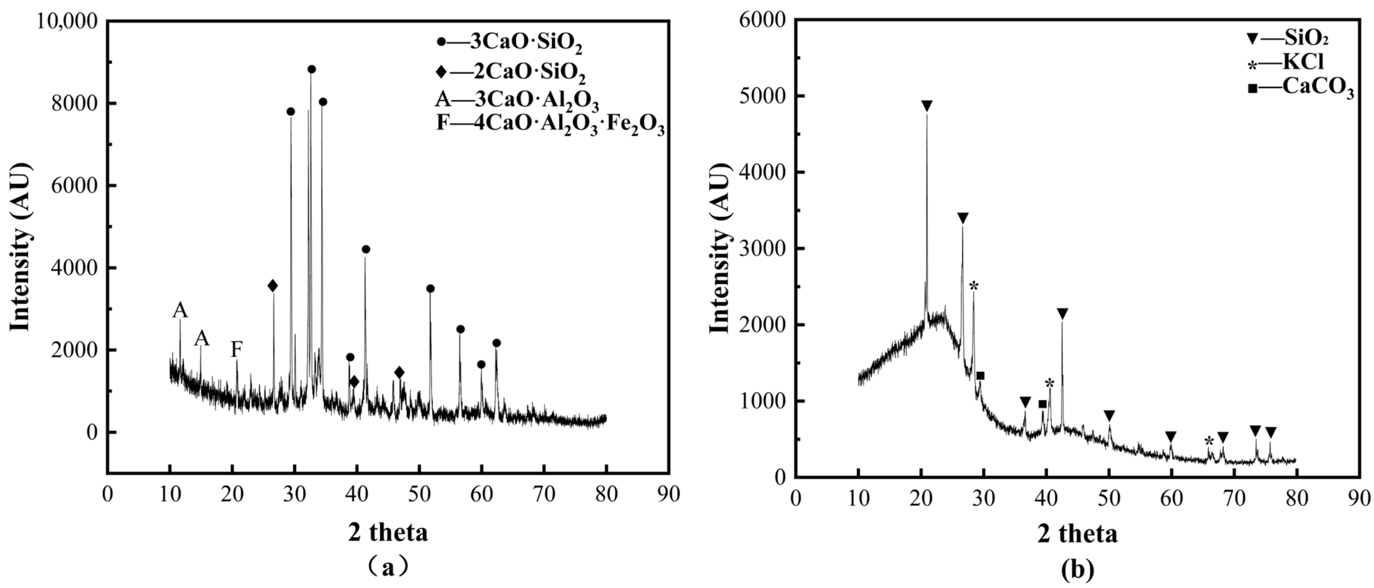

Figure 8. XRD diffractograms of raw materials: (**a**) cement; (**b**) CSA.

3.2. Slump

The slump test results are illustrated in Figure 9. The results indicate that the incorporation of both CSA and CF reduces the slump of concrete, with a decreasing trend observed as the dosage increases. Specifically, the concrete containing 10% CSA and 0.32% CF exhibits the lowest slump, representing a 65.2% reduction compared to the plain concrete. The reduction in slump caused by CSA can be attributed to two primary factors: firstly, the porous surface structure of CSA exhibits strong water absorption, absorbing a portion of the mixing water during blending and thereby reducing workability; secondly, the irregular particle morphology of CSA increases the frictional resistance within the mixture, further decreasing fluidity. Beskopylny et al. [56] reported a similar finding: coffee grounds biochar waste particles possess a porous structure and require more water than cement particles. Consequently, the slump of concrete decreases as the dosage of coffee grounds biochar waste increases.

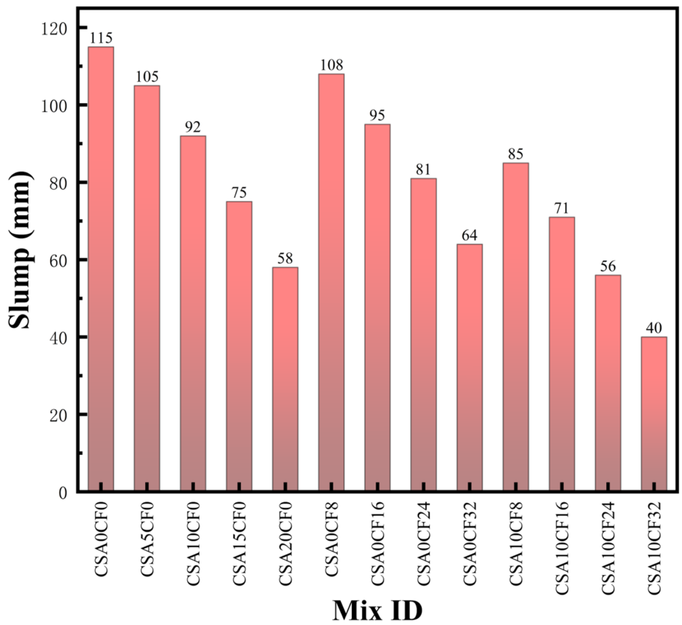

Figure 9. The slump of fresh concrete.

CF similarly absorbs water due to its porous structure [57], leading to a reduction in slump. Ahmad et al. [58] also attributed the reduced slump in sisal fiber-reinforced concrete to the fibers' absorption of a significant amount of water from the mixture. Moreover, when the CF content is excessively high, fiber agglomeration tends to occur, which further impairs workability. When CSA and CF are incorporated simultaneously, their combined effect on reducing slump may impact the practical applicability of the concrete. To ensure adequate workability, the dosage of superplasticizer can be appropriately increased to enhance the fluidity of the mixture to meet engineering requirements.

3.3. Density

Figure 10 presents the density of concrete with different mix proportions. The density exhibits a decreasing trend as the replacement ratio of cement by CSA increases. Compared to the plain concrete, the densities of CSA5CF0, CSA10CF0, CSA15CF0, and CSA20CF0 decreased by 2.5%, 4.4%, 6.2%, and 8.2%, respectively. This phenomenon is primarily attributed to the significantly lower bulk density of CSA compared to cement. Consequently, partially replacing cement with CSA results in a reduction in concrete density. This result is consistent with the findings of Perez et al. [59], where the incorporation of cane bagasse ash led to a decrease in the hardened density of concrete. Similarly, the density decreases with an increase in CF content. Compared to the plain concrete, the densities of CSA0CF8, CSA0CF16, CSA0CF24, and CSA0CF32 are reduced by 0.3%, 1.6%, 2.7%, and 4.7%, respectively. The reduction in density induced by CF is attributed to two aspects: on one hand, the low intrinsic density of CF; on the other hand, the incorporation of CF introduces additional pores into the concrete matrix, thereby lowering the overall density.

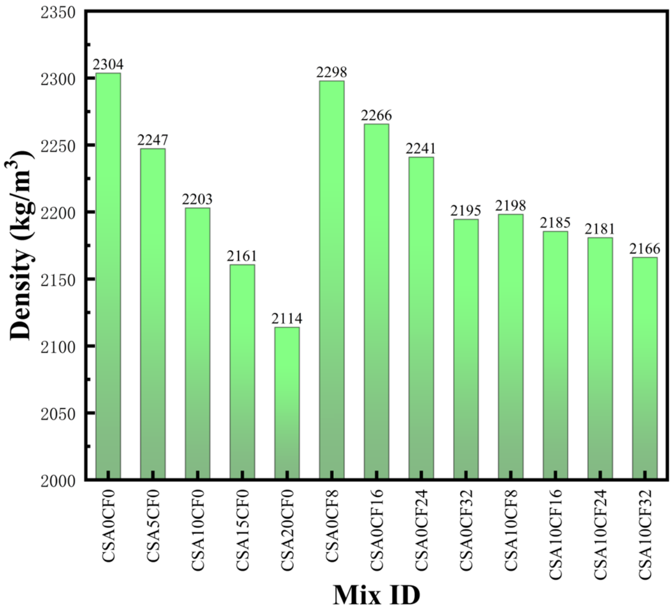

Figure 10. The density of hardened concrete.

3.4. Mechanical Behavior Characteristics

Compressive strength, flexural strength, and splitting tensile strength are critical indicators for evaluating the mechanical performance of concrete. The test results are illustrated in Figures 11–13. The results demonstrate that the compressive, flexural, and splitting tensile strengths of concrete with various mix proportions exhibit a consistent variation trend. For concrete with solely CSA, the compressive, flexural, and splitting tensile strengths decrease as the replacement ratio of CSA increases. The plain concrete exhibits strengths of 50.13 MPa, 5.73 MPa, and 4.14 MPa, respectively. When the CSA replacement ratio is 10%, the corresponding strengths are 43.7 MPa, 5.31 MPa, and 3.75 MPa, representing 87.2%, 92.7%, and 90.6% of the plain concrete, respectively, which still satisfies the strength requirements for C40 concrete. When the replacement ratio increases to 20%, the strengths further decrease to 37.74 MPa, 4.80 MPa, and 3.36 MPa, with reductions of 24.7%, 16.2%, and 18.8%, respectively. Özkılıç et al. [60] found that untreated pomegranate seed cake caused a gradual decline in compressive strength, whereas heat treatment (roasting) could mitigate this strength reduction. Therefore, future research could focus on the modification of CSA to alleviate the decline in mechanical properties.

For concrete incorporating only CF, the three strength indices initially increase and then decrease with increasing CF content. The peak strengths of 53.31 MPa, 6.53 MPa, and 4.45 MPa are achieved at a CF content of 0.24%, representing enhancements of 6.4%, 13.9%, and 7.5% compared to the plain concrete. It is noteworthy that the enhancement effect of CF on the flexural and tensile strength of concrete is greater than that on compressive strength. This is primarily because flexural and tensile failure is mainly caused by matrix cracking, allowing the CF at the crack sites to bear tensile stress and thereby improve flexural and tensile strength. In contrast, compressive strength is primarily derived from the aggregate and matrix; due to the limited crack width during compressive failure and the random orientation of the fibers, only a portion of the fibers can fully exert their bridging effect.

When the CF content is further increased to 0.32%, the strengths decrease to 50.93 MPa, 5.73 MPa, and 4.13 MPa, indicating that an appropriate amount of CF improves concrete strength, whereas excessive incorporation produces adverse effects.

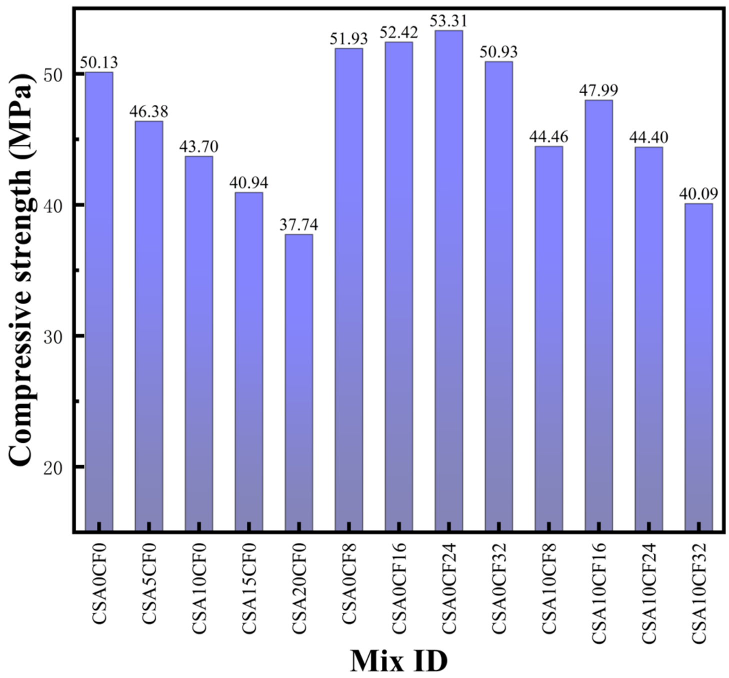

Figure 11. The compressive strength of concrete.

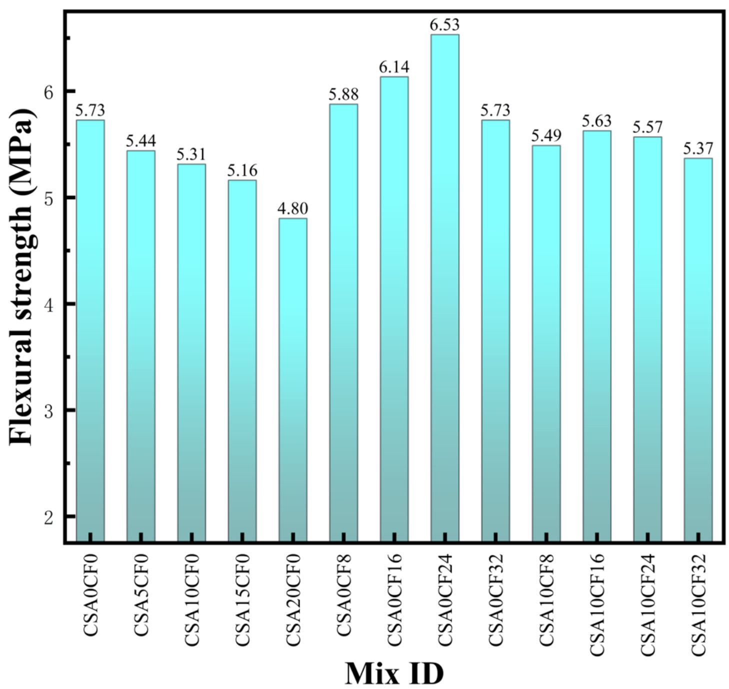

Figure 12. The flexural strength of concrete.

Under the same CF content, the strengths of concrete with both 10% CSA and CF are lower than those of concrete with CF only. However, the incorporation of CF mitigates the strength reduction caused by CSA. Kiamahalleh et al. [61] similarly found that incorporating sugarcane fibers into waste-based concrete improved its mechanical performance, with optimal properties achieved at a 3% fiber content. The concretes with both 10% CSA and

CF exhibit a similar strength trend to that with CF only, but the fiber content corresponding to the peak strength decreases from 0.24% to 0.16%. The peak compressive, flexural, and splitting tensile strengths are 47.99 MPa, 5.63 MPa, and 3.99 MPa, respectively, which are slightly lower than those of the plain concrete. These results indicate that the rational use of CSA and CF can simultaneously achieve the goals of maintaining mechanical properties and utilizing waste materials.

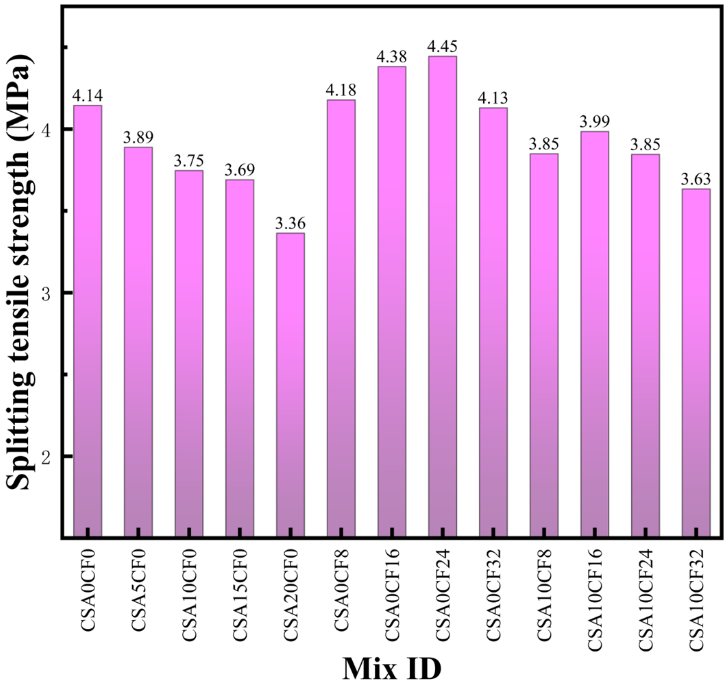

Figure 13. The splitting tensile strength of concrete.

Figures 14–16 illustrate the failure modes of specimens under compressive, flexural, and splitting tensile tests, respectively. The plain concrete and mixtures with solely CSA exhibit typical brittle failure characteristics: the compressive specimens fragment into loose pieces after testing, while the flexural and splitting tensile specimens fracture completely into two halves along a single dominant crack. In contrast, all specimens incorporating CF maintain integrity after failure due to the bridging effect of the fibers, indicating a transition in failure mode from brittle to ductile. Saad et al. [62] also discovered that the incorporation of banana fibers and palm leaf sheath fibers effectively improved the brittleness of high-strength concrete.

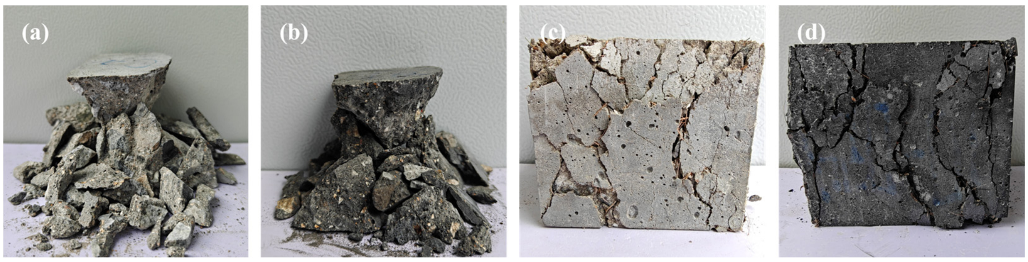

Figure 14. The specimens after the compressive strength test: (**a**) CSA0CF0, (**b**) CSA10CF0, (**c**) CSA0CF24, and (**d**) CSA10CF16.

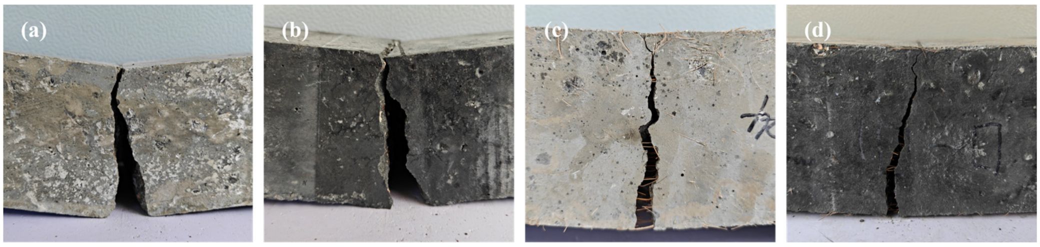

Figure 15. The specimens after the flexural strength test: (**a**) CSA0CF0, (**b**) CSA10CF0, (**c**) CSA0CF24, and (**d**) CSA10CF16.

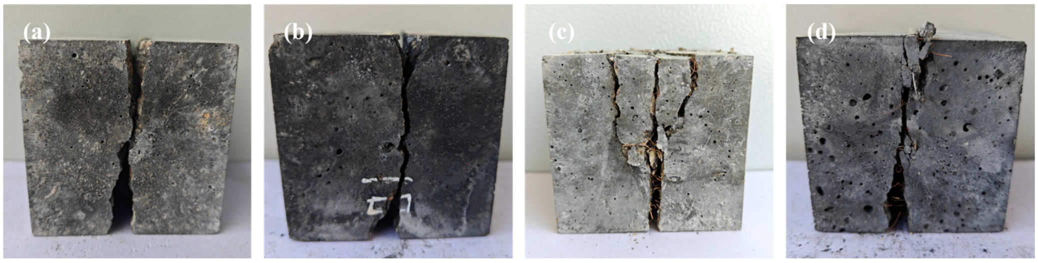

Figure 16. The specimens after the splitting tensile strength test: (**a**) CSA0CF0, (**b**) CSA10CF0, (**c**) CSA0CF24, and (**d**) CSA10CF16.

3.5. Mechanical Mechanism

Figure 17 displays the characteristics of coarse aggregates on the fracture surfaces of different specimens. Research indicates that the interfacial transition zone (ITZ) is one of the critical factors influencing concrete strength [63,64]. In specimens with CSA replacement ratios below 10%, the coarse aggregates at the fracture surface were mainly subjected to fracture failure (Figure 17a), indicating that the bond strength between the matrix and the aggregate remains sufficient, causing the failure path to pass through the aggregates themselves. However, when the CSA replacement ratio increases to 15–20%, the aggregates on the fracture surface predominantly show debonding from the matrix (Figure 17b), suggesting that a high CSA content significantly weakens the interfacial bond strength, rendering the ITZ the weak link in failure. Scanning electron microscope images further corroborate this finding. The incorporation of CSA exerts a detrimental effect on the ITZ: the ITZ in plain concrete is relatively dense, with hydration products well-encapsulated and bonded at the interface (Figure 18a). In contrast, the ITZ in concrete with 10% CSA is noticeably wider and looser, with reduced interfacial continuity (Figure 18b). This deteriorated ITZ cannot provide effective adhesion, leading to premature interfacial failure under stress, which macroscopically manifests as aggregate debonding and a sharp decline in strength.

The reduction in strength of CSA concrete is attributed not only to ITZ deterioration but also to a decline in the intrinsic bearing capacity of the matrix. The incorporation of CSA induces significant changes in the microstructure of the matrix. The matrix of plain concrete is dense with few pores (Figure 18a), whereas the matrix with 10% CSA exhibits a marked increase in voids, reduced compactness, and more internal defects (Figure 18b). This is primarily because the pozzolanic activity of CSA is relatively limited, resulting in less C-S-H gel generated from its reaction with the cement hydration product $Ca(OH)_2$. Furthermore, numerous unreacted particles are observed in the SEM image of CSA10CF0; these particles fail to exert effective filling or cementitious effects and instead act as defects and stress concentrators within the matrix, further exacerbating structural deteriorates. The increase in voids and decrease in compactness jointly lead to a reduction in the bearing capacity of the matrix. This explains why the strength of CSA concrete continues to decline even in the absence of aggregate debonding.

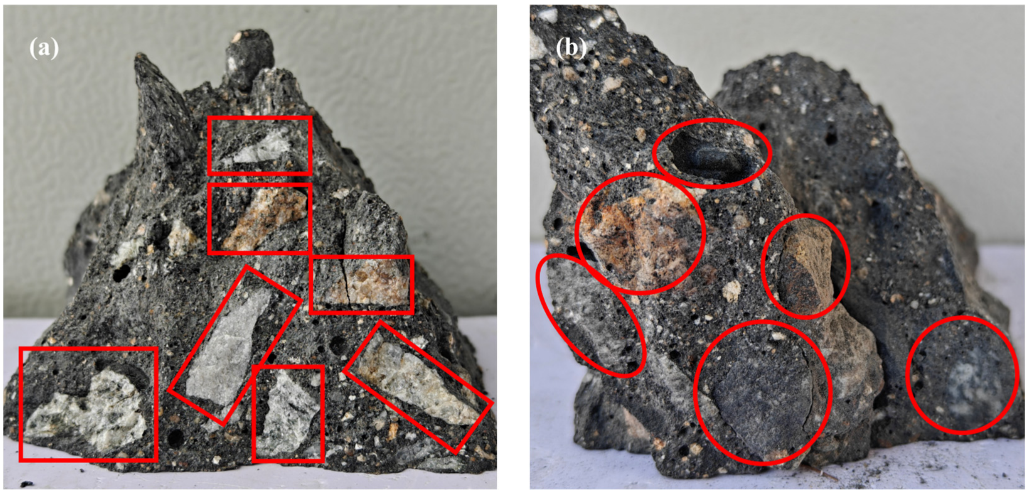

Figure 17. Features of coarse aggregates in fracture surface: (**a**) CSA10CF0; (**b**) CSA20CF0.

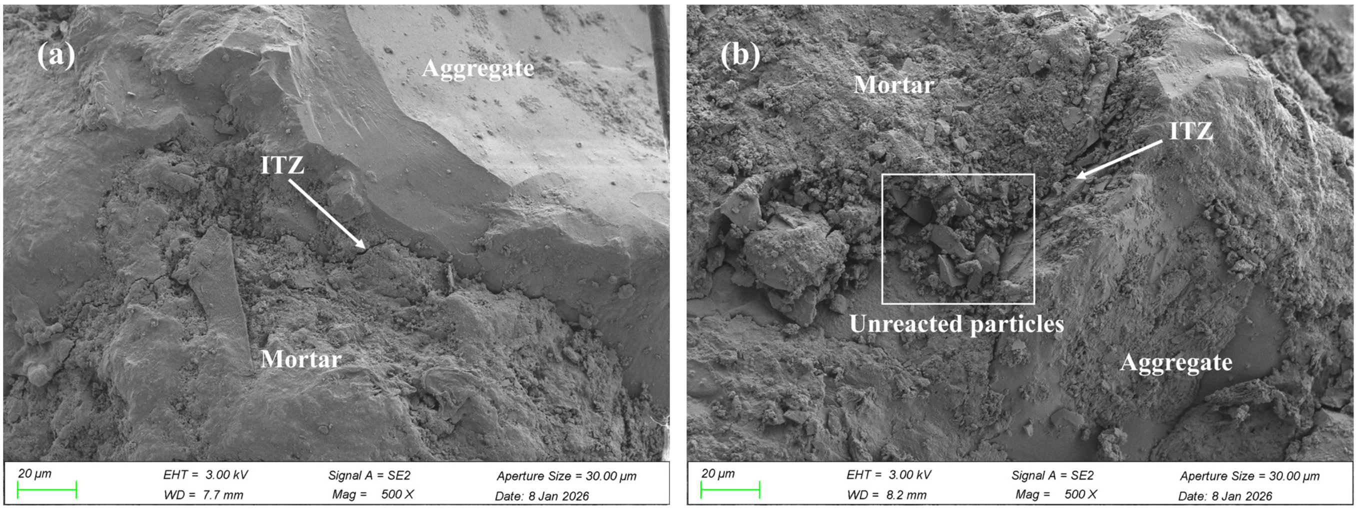

Figure 18. Microstructure of the concrete: (**a**) CSA0CF0; (**b**) CSA10CF0.

The uniformity of fiber distribution significantly influences the performance of CF-reinforced concrete. Figure 19 presents the distribution of CF within the fracture zones of different specimens. In concrete with CF only, the fibers are uniformly dispersed at CF contents of 0.08–0.24%, forming an effective bridging network. However, when the CF content increases to 0.32%, significant fiber agglomeration occurs, which not only disrupts the continuity of the stress transfer interface but also creates local weak zones, thereby weakening the fiber reinforcement effect and leading to a decline in mechanical properties. At the same CF content, the fibers in CSA0CF24 (Figure 19c) are evenly distributed, whereas obvious fiber clustering is observed in CSA10CF24 (Figure 19g). This discrepancy is primarily attributed to the reduced fluidity caused by CSA incorporation, which exacerbates the difficulty of fiber dispersion during mixing. Non-uniform fiber distribution restricts the effective bridging and stress transfer of CF, resulting in a lower fiber content corresponding to peak strength in the hybrid concrete.

Figure 19. Distribution of CF in fracture zone: (**a**) CSA0CF8; (**b**) CSA0CF16; (**c**) CSA0CF24; (**d**) CSA0CF32, (**e**) CSA10CF8; (**f**) CSA10CF16; (**g**) CSA10CF24; (**h**) CSA10CF32.

The interfacial bonding between the fiber and the matrix directly affects the strengthening efficacy of CF, and a sound interface is the prerequisite for effective stress transfer and fiber bridging [65]. Figure 20 shows the microstructure of CF-reinforced concrete. As shown in Figure 20a, the CF surface is rough, ensuring a tight bond with the cement matrix. This strong interfacial adhesion allows stress to be effectively transferred from the matrix to the fibers under tension, thereby fully mobilizing the bridging and crack-arresting effects of CF and significantly enhancing mechanical performance. However, in concrete with 10% CSA, the reinforcement efficiency of CF is limited by the deterioration of the ITZ induced by CSA. The SEM image in Figure 20b reveals that the ITZ between the CF and the matrix in the hybrid concrete is looser than that in concrete with CF only. The weakened interfacial bond restricts the effectiveness of fiber bridging, causing the optimal CF content in the hybrid concretes to decrease from 0.24% in concrete with CF only to 0.16%, and the peak strength is lower than that of the specimen with CF only. Nevertheless, the incorporation of CF still compensates for the strength loss caused by CSA to a certain extent, resulting in overall superior mechanical performance compared to concrete with CSA only.

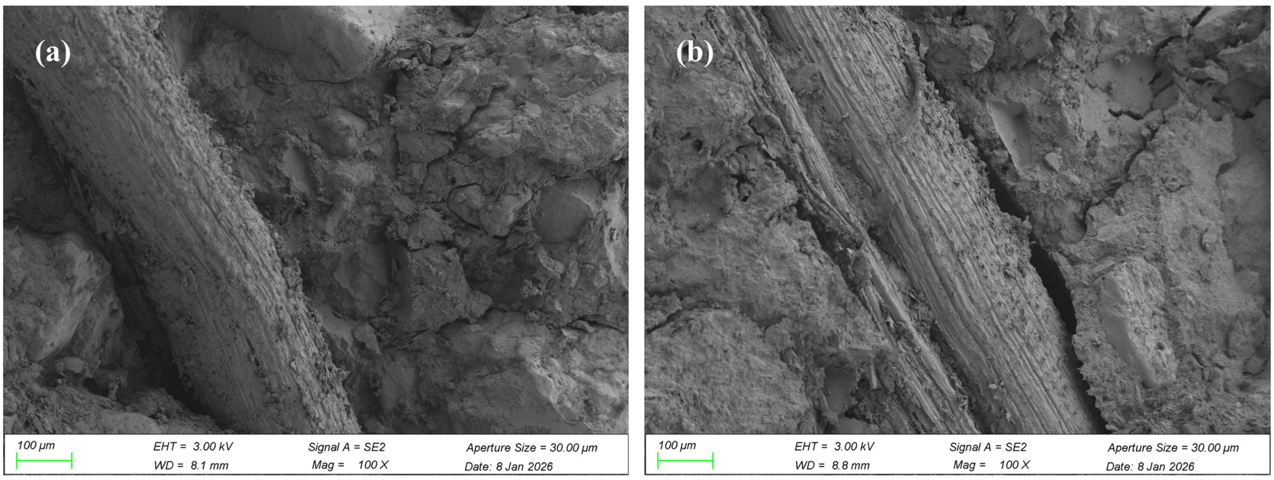

Figure 20. Microstructure of the concrete: (**a**) CSA0CF16; (**b**) CSA10CF16.

4. Conclusions

This study systematically investigated the effects of partial cement replacement with coconut shell ash (CSA) and the incorporation of coconut fiber (CF) on the slump, density, compressive strength, flexural strength, splitting tensile strength and microstructure of concrete. The main conclusions are as follows:

1. The incorporation of CSA reduced the slump and density of concrete. Furthermore, compressive, flexural, and splitting tensile strengths decreased continuously with increasing replacement level. However, the concrete with a 10% CSA replacement still met the strength requirements for C40 concrete.
2. The incorporation of CF significantly improved the mechanical properties of concrete, with an optimal dosage of 0.24%. At this dosage, compressive, flexural, and splitting tensile strengths increased by 6.4%, 13.9%, and 7.5%, respectively. Moreover, the failure mode shifted from brittle to ductile.
3. In the hybrid concretes, the combination of 10% CSA and 0.16% CF effectively compensated for the strength loss induced by CSA, achieving mechanical performance comparable to that of plain concrete. Due to the reduced workability caused by CSA and its impact on fiber dispersion, the optimal CF content in the hybrid concretes decreased from 0.24% to 0.16%.
4. Microstructural analysis revealed that the incorporation of CSA deteriorated the ITZ and increased matrix porosity, leading to strength reduction. CF mitigated crack propagation via a bridging effect; however, its reinforcement efficiency was constrained by the decreased interfacial bond quality resulting from CSA.

5. Limitations and Future Recommendations

The current study only conducted short-term performance testing under standard curing conditions and did not carry out durability tests such as impermeability or carbonation resistance. The alkali–silica reaction risk assessment or chloride permeability testing was not conducted to determine the impact of high K^+ and Cl^- levels in CSA. Furthermore, a more detailed quantitative analysis of concrete properties (such as rheology and porosity) is lacking.

The following are some recommendations for future studies:

1. Future studies should conduct durability tests on CSA and CF concrete to comprehensively evaluate the impact of CSA and CF on concrete properties.
2. The preparation process of CSA could be optimized, or pre-treatment methods could be explored to enhance its pozzolanic activity.
3. A lifecycle assessment and cost–benefit analysis should be performed to determine the overall environmental and economic impact of coconut shell waste concrete.

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